

1.1 description how we can solve this problem

The application allows customers to view their purchases quickly and shows crucial product details, such as production dates and, most importantly, expiration dates. In addition, it will help customers after buying the products by reminder them of expiration date. And it helps supermarkets in the stocktaking process, saving time and effort and minimalizing wasting products and throwing them away.

1.2 Identification of the information needs

We have defined some of **business role** in our Database:

- 1- PURCHASE may consists of many ITEMS, and it may consist of only one ITEM.
- 2- CUSTOMER may buy many ITEMS, or he/she may buy one ITEM. All the customer ITEMS are PURCHASE.
- 3- CUSTOMER may submit any number of ORDERS, each ORDER must be submitted by exactly one CUSTOMER. and CUSTOMER may not have submitted any ORDER.
- 4- ORDER must request one (or more) PURCHASE ITEMS, A given PURCHASE may not be requested by any ORDER or may be requested on one or more ORDERS.
- 5- The SCANNING must scan every PURCHASE ITEMS.
- 6- MARKET may deliver many PURCHASEs, and it may not deliver any PURCHASE.
- 7- TRACKING must remind the CUSTOMER of expiration date.